DB Final 2025 - OCR Extracted Text

Page 1

Mansoura {

Faculty of ¢ A eeeity j

SUN mn s (Model! |

Information oe x Informat Database s> ss ;

No. of SYSteom Depart BHON Science Final Exam, 2

—-0f pages 4 pages Ment 17/6/2025, Time: 120 minutes

isda Model | Total Marks: 60 aid es

Answ er 5 ‘Zi

setae All questions (Tot;

O1 — Otal Marks: 60)

., Choose the corre

“et answer [50 Marks|

. \ table is m INE iy

) it ha

. 2NF eliminates

If a table has'k Compose “ kev tA. BI. and a non-key attribute C depends only on A, thi

z Violates ° —

3) é NI ICN

Which normal form si non-prime attribute is transitively dependent on a super kev?

4 ve

If A>B and B=C a) FDs then A Be

5.

ble with columns [OrderID, Produ {) ty, ProductNat ely =

vi ut {

- ily on ProductiID

lication of data at several places is called Data

: b) Consistenc | \ i edu inc

\ functional dependency is 4 relationship between or among:

1) Ta) rows ©) relations d) attributes

Ifa relation fails to satisfy the n'* normal form condition, the highest normal form will be

; n-| ce) n+l d) nt2

Which of the following is a key characteristic of a distributed database?

f- ¢) No network d) Data splitting and

a) Centralized control b) single-point failure Hes ; ee ~

connectivity required replication

. Which architecture allows each node to have its own CPU, memory, and disk?

a) Shared Everything b) Shared Memory ¢) Shared Disk d) Shared Nothing.

Which of the following is a key advantage of distributed databases? a

a) Single-point failure b) eal Improved c) No need for a) Lower ala

availability network connectivity COsts

What is a drawback of naive partitioning? .

13- ae

a) High scalability b) Automatic c) Uneven resource d) WN

. ° > AG *

replication usage -overhe

Page 2

Consistens hanks

. | ‘ashing helps ‘at <

A) cry Pt dat: 2 “

nodes ‘A actoss b) Minimize data ¢)

What is the me —— e

4 : role of a "coordinator" in centralized transaction mama

a) Executes ; a

locally cutes all queries 6) Routes queries c) Stores all @at®” gy Mama

randomly redundantly

a Vertical Partitioning splits a table by: 4) Replication factors

a) Rows b) Columns c) Geographic regions

a Which partitioning Strategy maps keys to a logical ring? ay Naive Partitioning

a) Range Partitioning b) Hash Partitioning —_c) Consistent Hashing

A replication factor of 3 means:

oo Bceated 3 > correct

18- a) Data is stored on 3. b) Data is encrypted 3 c) Queries are — il

nodes times i

Parallel databases differ from distributed databases in that they or Avoid partitioning

“ < ssume .

19- a) Use unreliable b) Connot scale c) = Assume entirely

public networks horizontally Vw

ee a 4 9

What is a database transaction? aA. tipe dt database

| 20- + ole SQL query») A logical unit of work c) A backup operation index

a) A single : Aer

following is NOT a problem caused by concurrent transaction execution?

Which of the ol Aaiet RE DS d) Incorrect summary

n- Lost update problem — b) Dirty read problem SV OSOMTUS IES BRS UIE problem

apie pike A" in ACID properties stand for?

m What pe b) Aggregation c) Accessibility d) Atomicity

i a) Ava me purpose of the system log in transaction recovery?

what ® » user b) To record all ¢) To o timize SE NG :

to transaction operations er fc ey) Ta replicate data

23- a) ids : performance _ across nodes

pas property ensures that a transaction's changes are permanent once mene

wit? Ae b) Consistency ; i muted?

nicl? 2 ¢) Isolation a Dieskin

wa yh i operation ensures that log entries are written 10 disk before F ae

whit b) Undo : re committing?

ed? : : ©) Force-write

25- 4) ng to the following relations RI, ang R2. wi z d) Roll

coo qstance of R41 is related to more than » With Attributes RIA are

| ee One instance of ee B,C, D} ana RX y -

nein B,C, D} and b) RICA, B ¢ * So, the new nha Y, Z},

26- ni ZA}, R2AX,Y, 2)' D) and) RAB ns must

wv o—— integrity requires the fore: and REX Pa ON Dx} d)

a7 wets those that eon. ferentigs ~~ o Match the pri = RX y ra A i)

Pe ies sist ofa hicrarep, Entity rei nMary key Value . I

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Page 3

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Atibase Selena all design o ba inne

Ina base tens Unt

%- a bane relation, no attrib int

ule of a prim k fer to

a)Primary : ae thes :

b)Referentiat

In EER, th .

we > "

nS | existence of at least one shared sul

8) union ;

It is a collecti database

32- collection of Programs that enable users t pd maintata datao®

Te

AYDBA BYACIS

‘ i elation(s)

33. ‘According to Figure 1, the result of mapping this ERD produc ‘

A)t b)2

34. ——— are responsible for authorizing access to the database ar

a)DBMS b)DBA y) us for it at the

er asks for its

A virtual relation does not have to be in the database bot can be made if a 4m

35. time of the request.

a)Base relation b)Snapshot Query tates

. th attribute

Mapping the ERD model in Figure 1 produces a relation oumed PERSON with

36- cy ijt

a)5 by

atabase?

Which of the following ts ty pically NOT considered a characteriste of a database

37- a)Self-describing nature b)lsolation of data An Wade vith application independen

ons 4 generate

ER-to-Relational Mapping for Many-to Many relationships gen

t tional relatior y additional at

38 a)ignored Adasts ' :

Le ANSU/SPARC model, the schema deals with the user

F > sternal)) Natura: atte od

B have a relation (R) with a list of attributes (A, B,C, D, BF.

' Go, the set (A, B,C, D} is considered

: ' "

ls Rive key) Primary }) Candidate k

— Which data model organizes data into tables with rows and columns?

" >) My a)yNetwork mode biHicrarchical mod Relational mod Obs ri

? An Entity-Relationship (ER) model is used for:

4 a

: eee a) = Managing user b)Physical Jatabase cc) Writing SOL queric {)Kepresenting data at

| design the conceptual level

Ms. Which of the following is a component of an ER diagram? P

a)Entities b)Columns ©)Rows d)Indexes

Which ¢ tno ts. acai ¥

of the following is a disadvantage of file-based systems? .

44 a\Data ehariny b)Data redund ce _

ata redundancy ¢)Program-data d)Efficient quoty

independence processing |

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PAs 7

Page 4

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is. Uhe “NSS

4)3 ¢ ;

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a)| *YS, haw MARY can he designated as p oa

b)3 eM eS

Aceorgiy \ any!

49 RO Tabi Lown; a ‘ ‘ i idered the primary er...

wx hich Of the following attributes is cons:)None of them ' "

bY eZ

sp. As Shown jy Tabje ANC oe 1.2) onan @All of them

UP ritary key b)Candidate key one

Table 1

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ey Cae 31 2 | 43

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S following gue

)2- Answer the folic . lisgram

Q2 iag ERD diagr ¥ 2.5 Mark

ing the Follow A queries. (+x :

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p Find the number laced orders worth more than

>. Get customers wt er been ordered.

¢ ~ that have never be 3

' 1. Find products that

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